

Local Pachamama Beliefs and the Expansion of Cooperation among the Amazonian Kichwa (Pre-Registration)

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1. Background

1.1. Theory

For generations, thinkers have probed the relationship between religion and cooperation, variously construed (Durkheim, 2001; Malinowski, 1992; Rappaport, 1999). Recently, a newer body of research attends to the role that gods' knowledge, punishment, and moral concern may play in cooperative relationships. Specifically, some posit that there should be a positive relationship between cooperation and the degree to which gods are believed to know and care about, and punish, antisocial behavior (Johnson, 2015; Norenzayan, 2013). One variant of this theory posits that humans evolved a sensitivity toward supernatural punishment because it reduces the costs of secular punishment for engaging in self-interested behavior (Johnson, 2005). Another variant posits that, because of the prosocial effects of such beliefs, traditions that espouse them will attract followers, thus facilitating broader cooperation, in turn allowing some groups to outcompete others in a cultural evolutionary scenario (Norenzayan et al., 2016). This cycle may thus partially account for the global dominance of the so-called "prosocial religions" such as Christianity and Islam. One implication of this theory is that, while such prosocial religions contribute to the expansion of cooperation, the cooperation promoted by smaller, "indigenous" or "local" traditions is presumably restricted to local communities. In other words, "world" religions such as Christianity and Islam may scale up cooperation to very wide scales and thus make it largely anonymous, whereas "local" traditions are exactly that: constrained to facilitate local, small-scale, and primarily face-to-face interactions.

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The empirical study of the relationship between religion and cooperation is generally mixed (Billingsley et al., 2018; Gomes & McCullough, 2015; Shariff et al., 2016; Van Elk et al., 2015), but the specific work on culturally postulated beliefs and cooperation is more robust. While cross-cultural work shows that beliefs in supernatural punishment and the breadth of gods' knowledge are positively associated with fairness and generosity beyond parochial boundaries (Lang et al., 2019; Purzycki et al., 2016), the degree to which gods are thought to care about morality is not (Bendixen et al., 2023). In fact, across individual- and group-level studies, the association between gods and moral concern has been shown to be cross-culturally more common than previously claimed, extending even to small-scale societies in which cooperation is restricted to immediate groups (Lightner et al., 2023; Purzycki, 2011, 2025b; Purzycki et al., 2022; Singh et al., 2021; Townsend et al., 2020). Furthermore, previous studies could not compare so-called “moralistic traditions” whose gods are “specifically supportive of human morality” such as fairness and harm avoidance (G. E. Swanson, 1960) with traditions whose gods are more associated with other affairs, such as over-exploitation of resources, ritual, etiquette, and so forth (Purzycki & McNamara, 2016). Here, we examine the relationship between such “local” beliefs and cooperation among a community of Ecuadorian Kichwa.

1.2. Ethnographic Background: Lowland Kichwa

With a population of around 125,000, the lowland Kichwa (Quichua) are the largest of 11 indigenous minority ethnicities in Amazonian Ecuador (Heredia-R et al., 2020; Vasco et al., 2018). The ancestral Kichwa language is a member of the Quechua dialect continuum, with limited mutual intelligibility between lowland and high sierra Kichwa dialects within Ecuador (Grzech et al., 2019). Because of this, the lowland Kichwa have long been stereotyped as recent Andean migrants into the Amazon, but this claim is almost certainly false; comparative evidence instead indicates that the lowland Kichwa are the ethnic descendants of integration between precontact groups such as the Shuar, Achuar, and particularly Quijo in the wake of the Spanish conquest and the dissolution of the Incan Empire (Heredia-R et al., 2020; Uzendoski & Whitten, 2014). Nevertheless, assumptions about an Andean origin have frequently been used to deny Kichwa land rights in the Amazon (Uzendoski & Whitten, 2014). Today, Spanish use is rapidly increasing, with only a minority of elders remaining Kichwa monoglots and most younger and middle-aged people either monolingual in or preferring Spanish (Grzech et al., 2019).

The lowland Kichwa are strongly tied, economically and ideologically, to the Amazon rainforest. Kichwa tend to devote a smaller portion of their land to agriculture or pasturage than mestizo colonists or other Eastern Ecuadorian ethnic groups in the Amazon region such as the Shuar, despite having larger households and larger total landholdings on average (Vasco et al., 2018). Households thus often have an abundance of land dominated by unmodified rainforest (Davis et al., 2017). Market integration has increased since the 1970s, with more extensive road networks connecting Kichwa territories to the rest of Ecuador. Accordingly, the production of cash crops

such as cacao and coffee has grown in economic importance (Heredia-R et al., 2020). Oil extraction, an extremely politically and symbolically charged issue in the Ecuadorian Amazon, drives both market integration and breakdown of traditional economic and cultural practices (Bozigar et al., 2016). With increasing market integration, migration to urban areas is increasing, often to access education (Davis et al., 2017). Nevertheless, compared with other groups in the region, Kichwa tend to have lower levels of outmigration and longer durations of residence in or near their current dwellings (Vasco et al., 2018). People who depart ancestral villages for wage labor frequently return, further suggesting constrained geographic mobility (Davis et al., 2017).

The lowland Kichwa continue to practice a form of small-scale agriculture known as *chagra*, which is characterized by extremely high levels of crop and product diversity on small (0.05 to 3.00 hectare) plots attached to private residences (Heredia-R et al., 2020; Perreault, 2005). The *chagra* system has been practiced in this region for millennia and makes sustainable use of rainforest land that is poorly suited for more intensive agriculture (Heredia-R et al., 2020; Vera-Velez et al., 2019). Starting in the 1970s, many indigenous Amazonian peoples began to ranch cattle, largely driven by a need to preserve land titles against claims by in-migrating mestizo colonists and oil companies (Rudel et al., 2002). Nevertheless, *chagra* cultivation remains ubiquitous among the Kichwa (Heredia-R et al., 2020; Vera-Velez et al., 2019). Overall, even in a changing economic and cultural landscape, Kichwa in Napo and nearby provinces have comparatively moderate levels of market integration and stronger ties to particular localities than other ethnic groups in the region.

1.2.1. Kichwa Religion

Lowland Kichwa communities are largely Christian, with Protestantism and Catholicism varying in proportion across locales. However, the Kichwa approach to religion is syncretic, with traditional animistic beliefs and practices fluidly mingling with Christianity in daily life (Uzendoski, 2005; Uzendoski & Whitten, 2014). In line with Amazonian ethnic origins, traditional lowland Kichwa mythology and religion share more similarities with those of neighboring indigenous Amazonian groups than with other Quechua peoples (Grzech et al., 2019). Kichwa beliefs heavily focus on the life of the forest and on kinship among humans, as well as between humans and forest beings (Uzendoski, 2005). Complex, multipart wedding ceremonies are central events for the reinforcement and creation of these kin ties and help anchor a broader cosmology of kinship between humans and nonhuman spirits (Uzendoski & Whitten, 2014). A common theme in Kichwa religion is the interchangeability of human and forest spiritual essence, and indeed transformation in general (such as between plant and human “kin”; T. D. Swanson, 2009; Uzendoski, 2005). Shamans, called *yachaks*, are important (if sometimes morally ambivalent) figures in most communities and help navigate transformative relations between humans and spiritual powers, often in service of healing (Uzendoski, 2005; Uzendoski & Calapucha-Tapuy, 2012).

Plant products play crucial roles in the Kichwa cosmos. Ayahuasca, a potent hallucinogen with growing appeal for Western tourists, features in certain shamanic ceremonies (Naranjo, 1979; Salibová, 2020; Uzendoski, 2005). A spicy sauce made of *aji* peppers is used both in daily cuisine and for punishment of misbehaving children through shamanic administration to the eyes. Guayusa, a caffeinated tree leaf made into a tea, is particularly ubiquitous in daily and ceremonial life. An important morning ceremony, the *waysa upina*, revolves around the preparation and serving of guayusa tea by women (Suzina, 2023). *Chicha*, a fermented beverage typically made with yuca (manioc) or the savory tree fruit *chonta*, is also a symbolically potent presence at virtually every gathering (Tobes et al., 2022). *Mingas*, or work groups, are important focal points for Kichwa communities (Vasco et al., 2018), and participants can typically expect to be served *guayusa* and *chicha* by women at every *minga*. *Changra* gardens are used to grow manioc, *guayusa*, *chonta*, and other products that both physically feed and spiritually nourish the community. Strong symbolic links thus exist between women, gardens, and the central consumables in Kichwa ritual.

Over the course of colonization and modernity, these links and the centrality for Kichwa life of the rainforest and its products (the *Kawsak Sacha* [living forest]) have become assimilated to a pan-Quechua concept and deity, *Pachamama* (Minoia et al., 2024). This name, originally of highland Andean origin, roughly translates to “the Space-Time Mother” or, perhaps less evocatively, “the World Mother” or “Mother Earth” (Inquilla-Mamani & Chambi-Apaza, 2019; Kohn, 2013; Uzendoski & Calapucha-Tapuy, 2012). It is possible that deepening Pachamama beliefs have been subtly incentivized by contact with outsiders, particularly sympathetic Westerners who associate Pachamama with environmentalism and ecology. Indeed, our data (see below) reveal abundant conceptual links between Pachamama and ideals of protecting the environment, not polluting, and not cutting down trees. However, despite being an Andean deity grafted onto earlier layers of rainforest-focused spiritual beliefs, Pachamama has become a genuinely important personage in Kichwa cosmology. Roughly speaking, she summarizes or stands in for the entire web of relations that make up the *Kawsak Sacha*, the *runa*, and the spirits and powers that link them. For most lowland Kichwa, Pachamama therefore coexists unproblematically alongside the Christian God, the Virgin Mary, and countless other spiritual powers as a feminine personification of growth and land.

1.2.2. Yawari Village, Napo Province, Ecuador

Our field site was Yawari, a small (population ≈ 200) Kichwa village in the Rukullakta pueblo of Napo Province. Napo is the most populous province in the vast lowland Amazonian region that extends eastward from the high sierra of the Ecuadorian Andes. Its Kichwa residents describe themselves as *Napo Runa* (“Napo people”; Uzendoski, 2005). Rukullakta is an autonomous Kichwa pueblo with around 2,300 inhabitants across 17 villages and more than 400 km², most of

which is rainforest (Erazo, 2013; Varela, 2025). The pueblo sits a few kilometers northeast of the local canton town of Archidona and about half an hour's drive from the provincial capital of Tena. Rukullakta was formed in the late 20th century and incorporated in 2006 to assert local Kichwa land rights in the face of encroachment by mestizo colonists and mining companies (Erazo, 2013). Its charter mandate includes ecological protection of the rainforest as well as putting Kichwa land to sustainable economic use (Varela, 2025). Rukullakta villagers previously ranched cattle but now more commonly participate in cacao and coffee production to generate income, solidify land claims, and provide legal protection against oil mining on pueblo lands (Varela, 2025). Most households also maintain a *chagra*. A single-item question in our post-experiment interviews asked what percentage of food consumed in each participant's household the previous day was purchased at a store; the mean response was 51.1% (SD $\approx$ 30%), indicating only partial market integration, albeit with considerable inter-individual variation. Religiously, Yawari village is entirely, if often nominally, Catholic; there is no parish church, but a priest visits on Sundays to say Mass in the village community center.

The Napo region is a center of Kichwa culture, with around 56% of the province's residents claiming indigenous identity as of 2003 (Informe Social, 2003). Triumphal statues of Jumandi, a local Quijo hero killed in a symbolically important 16th-century uprising against the Spanish, stand prominently in traffic circles at the northern entrances to both Tena and Archidona. Relations between Kichwa and mestizo colonists are characterized by both prejudice and transactional pragmatism; many Kichwa live in the urban centers and work in professional jobs, and local mestizos have frequently assimilated to Kichwa customs such as drinking guayusa and eating *chontacuro* palm weevil larvae (Chimbo-Gándara et al., 2024). Nevertheless, cultural and class barriers ensure that Napo indigenous and mestizo identities remain separate. In a telling example, an oratory shrine for the recently canonized Catholic Venezuelan saint José Gregorio Hernández stands in a small jungle clearing just outside the main entrance to Yawari on the road to Archidona. Despite being less than 1,000 feet from the Yawari Community Center, the shrine attracts almost exclusively mestizo worshippers, and Kichwa informants in Yawari generally disdain the saint and his mestizo devotees.

Compared with non-pueblo-dwelling Kichwa, then, Rukullakta Kichwa may be particularly culturally insulated, with many residents only occasionally venturing to Tena or interacting with outsiders. Pueblo inhabitants take their orientation instead from the opposite direction, the forest and its symbolic history. In particular, the isolated mountain *Yawar Urku* (mountain of blood) rises from the forested lowlands near Rukullakta. The mountain was the site of a successful battle against the Spanish during the colonial era (Erazo, 2013). Kichwa in the pueblo, and especially Yawari, the mountain's namesake, strongly identify with *Yawar Urku* and consider it an important sacred site, frequently assimilating it to the (separate) historical Jumandi uprising. Hence, the Yawari village name evokes both the local sacred mountain and the concept of "blood," with resonances both of intergenerational kinship and historical struggle. Kichwa

identity in our field site is thus strongly oriented around symbolic touchstones of rainforest life and forceful assertions of Kichwa identity against encroaching outsiders.

1.2.3. Ethnographic Interviews

Our experimental and ethnographic methods required that we select two representative, culturally salient deities — one “moralistic” and one “local” — about which to ask questions and so ultimately examine the role of “local” beliefs on intergroup cooperation. To represent “moralistic” traditions, we needed to select a “moralistic” god as our target. We anticipated that the Christian God would serve as the moralistic deity in the game experiments in Yawari, but in the absence of extant research, it was unclear whether belief in Pachamama would be sufficiently salient for an adequate local god. It was also possible that Catholic saints, local Marian apparitions (which are plentiful), or more specific local or forest spirits would better serve in that role. We therefore conducted preliminary ethnographic interviews with Yawari residents ($n = 20$) to gather both quantitative and free-list data about local religious beliefs. These Religious Landscape Interviews — so called because they represent the first-sweep effort to understand local spiritual outlooks — were conducted in Spanish, as were all subsequent experiments and games.

Free-list interviews allow interviewers to gather naturalistic cultural knowledge that can be used to calculate cultural salience (Purzycki, 2025a). For example, the first item in our preliminary interviews asked subjects to simply “Please list all of the gods or spirits that you can think of that people believe in here [Yawari].” Subjects then ranked all the gods or spirits they listed in order of importance, then indicated whether each one (a) was concerned about how people behave; (b) could see into human beings’ thoughts and feelings (i.e., was psychologically omniscient); (c) punished people for bad behavior; and (d) rewarded people for good behavior. Salience for each deity was calculated using Smith’s S (Smith & Borgatti, 1997):

$$S = \frac{\sum(\frac{n+1-k_i}{n})}{N}$$

Here, k_i is the item order number, n is the total length of the list, and N is the sample size for the interview. Thus, if every single respondent listed the same god and also ranked that deity first in importance, S would = 1.0 for that deity. To generate uncertainty intervals for these salience estimates, each deity listed by subjects was modeled independently using an ordered beta-distributed model with uninformative priors (Major-Smith & Purzycki, 2026). Analyses were conducted in the AnthroTools package in R (Purzycki & Jamieson-Lane, 2017). Respondents listed 16 deities in total, but only the Christian God ($n = 15$), Pachamama ($n = 15$), Jesus Christ (in various formulations; $n = 7$), and the Virgin Mary (in various formulations;

$n = 6$) were listed by more than five respondents.² Salience analysis indicated that the Christian God and Pachamama were by far the most culturally salient deities (Figure 1). Analyses of follow-up items indicated that respondents viewed the Christian God as somewhat more punishing, omniscient, and likely to reward good behavior than Pachamama, though both deities were equally concerned about morality (Table 1). As such, we selected the Christian God as the “moralistic” god and Pachamama as the “local” god for the experiments and interviews.

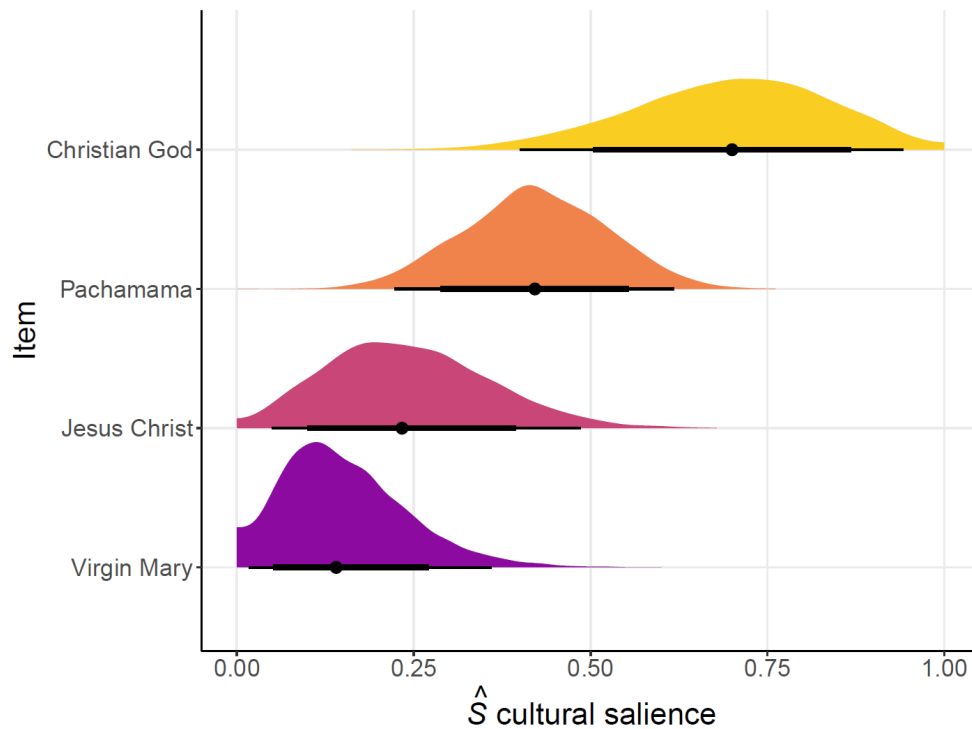


Figure 1. Cultural models of deities. Black circles are Smith’s S values, with error bars displaying 89% (thick lines) and 95% (thin lines) credible intervals. Deities listed by 5 or more of the 20 Religious Landscape Interview participants are included here.

² Pachamama has often been identified with the Virgin Mary in Quechua and Andean contexts elsewhere in South America (Damian, 1995), raising the question whether the two deities would be better treated as one overlapping category in our dataset. However, we found no evidence for this assimilation, and in fact numerous Yawari respondents informally — but forcefully — asserted that Pachamama was *not* the Virgin Mary, indeed seeming puzzled and possibly offended by the question. We concluded that Pachamama was validly treated as a distinct deity in our sample.

Table 1: Punishment, knowledge breadth, moral concern and reward of most salient deities, based on Religious Landscape Interviews ($n = 20$).

	Punishes	Sees	Moral concern	Rewards
Christian God	12/15 (80.0%)	15/15 (100%)	15/15 (100%)	15/15 (100%)
Pachamama	11/15 (73.3%)	10/15 (66.7%)	15/15 (100%)	14/15 (93.3%)
Jesus Christ	4/7 (57.1%)	6/7 (85.7%)	7/7 (100%)	7/7 (100%)
Virgin Mary	2/6 (33.3%)	3/4 (75.0%)	5/6 (83.3%)	5/6 (83.3%)

2. Methods

2.1. Theoretical Model

Our theoretical causal model is illustrated in Figure 2 (this model was adapted from our core causal model for the wider Gods, Games, and the Socioecological Landscape project, available [here](#)). Our focal estimand is the causal impact of beliefs in supernatural punishment and omniscience of the local deity (Pachamama) on impartial fairness and generosity towards distant co-religionists of the same local religious tradition (i.e., fellow Kichwa believers in Pachamama). We believe the following sociodemographic variables may potentially cause both Pachamama beliefs and extended religious cooperation (Albrecht & Heaton, 1984; Baimel et al., 2022; Purzycki & Bendixen, 2024; Scheepers et al., 2002; Vardy et al., 2022) and thus treat them as confounders of this effect: age, sex, years of formal education, current food security, and number of children. We also include “market integration” as an additional confounder. That is, Kichwa who are more reliant on *chagras* and hunted or foraged food may both have a stronger connection with the land than market-integrated neighbors, and hence stronger Pachamama beliefs, as well as perhaps a greater need for more expansive social and cooperative networks. Alternatively, more market-integrated individuals may be more aware of an abstract identification with anonymous Kichwa in a neighbouring province and connect that identification with Pachamama, whereas subsistence farmers may be more focused on their own locality and thus see distant Kichwa as less salient. Either way, market integration is a potential confounding factor.

Further, while game order and comprehension checks are not confounders (they cannot cause our exposure), we included them to minimize uncertainty around the focal estimand, given that they likely explain variation in cooperative behaviour. Because participants are assumed to not regularly interact with distant co-religionists, we assume that participants’ beliefs — which are the end result of years of enculturation and individual processes — are not the result of their behavior in these games, minimising the risk of reverse causality bias. Note, however, that we

cannot rule out prior cooperative behaviour outside these games causing both subsequent religious beliefs and cooperative game behaviour. We must also assume that there is no unmeasured confounding (denoted by the node ‘ U ’ in Figure 2), measurement error, or selection/collider bias in order to satisfy the requirements of causal inference (Hernán & Robins, 2020).

In addition to the core confounders introduced above, other variables may plausibly confound our target estimand. These include Catholic religious beliefs and rituals and engagement in Pachamama rituals. While each of these variables may plausibly cause cooperation with distant Pachamama co-religionists, given the cross-sectional nature of the data it is impossible to know whether these variables cause Pachamama punishment and omniscience beliefs (i.e., they are confounders, and need to be adjusted for to remove bias), or whether they are *caused by* Pachamama beliefs (i.e., they are mediators of our focal estimand, and do *not* need to be adjusted for when estimating our causal effect of interest). Causal effects in both directions are also possible; hence the bidirectional arrows between these variables in Figure 2. For instance, stronger beliefs regarding Catholicism and participation in Catholic rituals might result in weaker beliefs regarding Pachamama; equally, stronger beliefs regarding Pachamama might result in weaker Catholic beliefs and practices. Or, perhaps beliefs in both stem from some common underlying latent or unmeasured factor, such as overall religiosity or other cognitive faculties. Similar bidirectional effects are also plausible for engagement in Pachamama rituals (i.e., it could cause, and be caused by, Pachamama beliefs). Given this ambiguity, using a single model to estimate an unbiased causal effect is impossible, and instead sensitivity analyses of multiple statistical models corresponding to different causal data-generating mechanisms are required (Major-Smith, 2023). These will be discussed in more detail below.

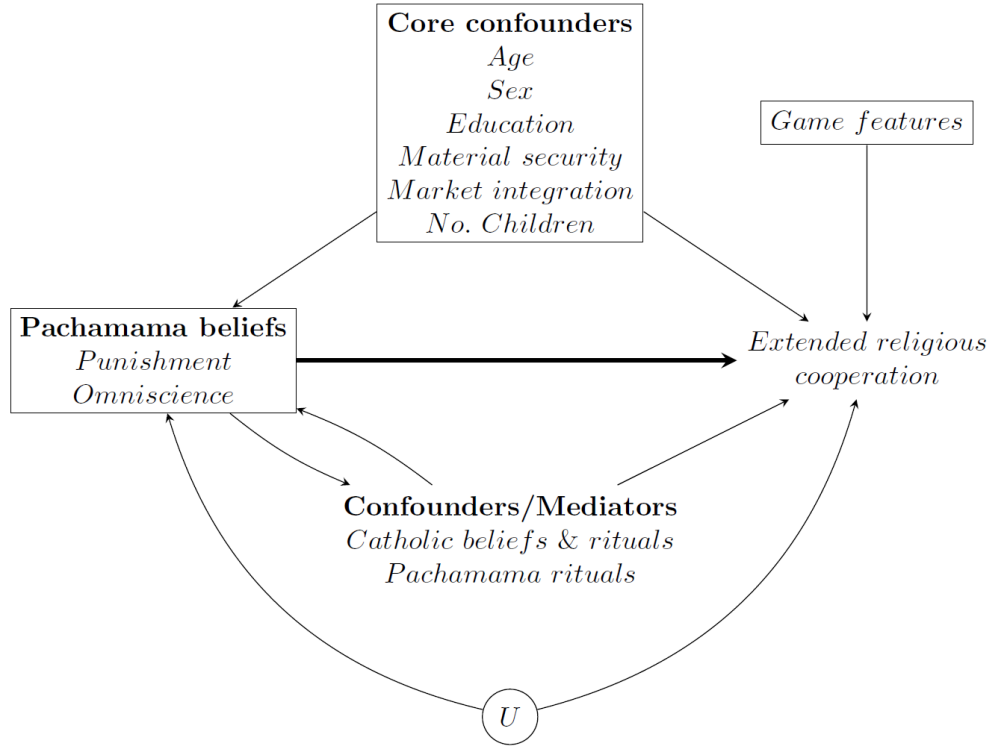


Figure 2. Causal graph displaying our core causal assumptions regarding the data-generating process.

2.2. Materials and Procedure

2.2.1. Demographics

We asked a battery of standard demographic questions. Most relevant to this project are the core confounding variables of age, sex, years of formal education, current food security, number of children, and market integration discussed above. For years of formal education, we simply asked for all years of education in a formal school setting, and for food security, we asked whether or not participants worried about getting enough food to eat over different time horizons (month/6 months/year/5 years; yes = 1; these four questions will be averaged together). For market integration, participants were asked the proportion of food purchased at a store, both for the preceding day and the past month.

2.2.2. Beliefs

Our focal exposures are supernatural punishment and omniscience of Pachamama, which were measured as part of a larger religiosity survey, separate from the Religious Landscape Interview, assessing beliefs and practices towards both the Christian Catholic God (our “moralistic” god) and Pachamama (the “local” god). We measured punishment using the mean of two questions:

(1) Does **Pachamama** ever punish people for their behavior? (yes = 1, no = 0) and (2) Can **Pachamama** influence what happens to people after they die? (yes = 1, no = 0). To measure Pachamama's omniscience/knowledge breadth, we took the mean score of: (1) Can **Pachamama** see into people's hearts or know their thoughts and feelings? (yes = 1, no = 0) and (2) Can **Pachamama** see what people are doing if they are far away, in Guayaquil (Ecuador's largest city, 270 km away on the Pacific coast)? (yes = 1, no = 0).

The same questions as above were used to assess punishment and omniscience beliefs regarding the Christian Catholic God (with "Pachamama" replaced by "Dios" [God]). For Pachamama and Catholic rituals, we used the question "How often do you perform rituals or ceremonies devoted to [deity]?", answered on a five-point Likert scale (0) Very rarely/Never; 1) A few times per year; 2) A few times per month; 3) A few times per week; 4) Every day or multiple times per day).

2.2.3. Cooperation

To measure impartial fairness and generosity, we deployed variants of the Random Allocation Game and the Dictator Game. In one variant – the Self Game – participants allocated money between themselves and an anonymous, geographically distant co-religionist from Palimbe, a randomly chosen Kichwa village in the neighboring province of Pastaza. Palimbe lies deep in the rainforest, a significant distance along poor roads from Pastaza's provincial capital of Puyo. No one in Yawari had heard of Palimbe or claimed to know or be related to anyone who lived there. The other variant – the Local Game – entails participants allocating money between a co-religionist from their own community and another anonymous co-religionist from Palimbe. "Co-religionists" here were explicitly identified as fellow Kichwa believers in Pachamama. Participants played the Random Allocation games first (counterbalanced across participants) followed by the Dictator games (also counterbalanced).

In the Random Allocation Game (Hruschka et al., 2014), participants were given the option of two individuals — denoted by two labelled but otherwise identical cups — and asked to think silently of the individual to whom they wished to allocate a coin. They then had to roll a fair die with an equal number of black and white faces. If the die came up black, they had to put the coin into the cup they thought of, but if the die came up white, they had to put the coin into the *opposite* cup. In each variant of the Random Allocation game, participants allocated 30 \$1 coins in USD (the legal tender currency in Ecuador) for a total of USD \$30. The mean distribution to distant co-religionists in perfectly fair play would therefore be \$15 for each iteration of the Random Allocation Game. For the Dictator games, participants were given ten \$1 coins to divide between the two cups however they wished. Across two games, then, the total amounts were a maximum of \$80 (approximately an average daily wage).

2.2.4. Game Features

As mentioned above, we also include game order and comprehension checks as factors that might affect the outcome, but we assume these have no bearing on beliefs and demographics. The game order was randomized across participants, but we anticipate that playing the Self games second, for example, might induce a little more self-interested behavior. Game comprehension checks were designed to screen out individuals who did not understand the game (and hence did not play these games), but even among those who passed, the number of comprehension checks needed may have impacted game behaviour (assuming there was variation).

2.2.5. Procedure

To minimize collusion, we conducted all experiments first and followed up with demographic and religiosity interviews over the course of the following week. Data collection was conducted in the local school in a single day, in the center of Yawari village. Subjects played games alone in temporary tarp booths in classrooms facing a wall. The school grounds were walled, with only one entrance and exit to the street. The game area was part of a long “L” of classrooms enclosing a central outdoor auditorium. The only way to move between classrooms was via an outdoor concrete walkway that connected to the auditorium/gym space. Participants gathered in the auditorium area to wait their turn to play, and barriers of tarp were set up to prevent visual access to the walkway and classrooms. We provided snacks, drinks, and lunch in the waiting area and, throughout the day, called participants one at a time to exit the waiting area via an opening that faced away from the classrooms. They then were led via the walkway to a classroom booth, given an explanation of the games by a research assistant, tested for comprehension, and allowed to play. Trays of coins and games were prepared in a separate classroom and were shuttled out to the booths by research assistants, who also collected completed game trays for counting in the preparation room. Participants were given numbered ID cards with randomization codes for game order and were told to keep the cards for the interviews the subsequent week. Upon completing the game, participants were escorted to the school exit in the same direction as the flow of participants from the waiting area. All movement from the waiting area to game booths to exit was thus one-way. At the exit, participants signed out with a volunteer. Signing out entailed selecting an interview slot for the upcoming week. Participants were not allowed to reenter the school grounds after departing and were told not to talk about the game with anyone. No further participants were allowed into the school grounds after game play began in mid-morning. Participants in the waiting area were told not to discuss the game before playing. Phone use was not allowed in the waiting area.

One participant, a day laborer who would be unavailable the following week, carried out the demographics and post-game interviews on the day of game data collection in a classroom after

other players had departed. All other interviews were conducted in daylong blocks over the ensuing week in the neighboring community center. Participants arrived on the day designated by their signup slot and were interviewed in different parts of the community center, with between four and five assistants conducting interviews in parallel. Food and drinks were provided. After completing interviews, subjects were paid the flat rate for their participation (Section 2.4) alongside their self-allocations from the Random Allocation and Dictator Self games, then signed out.³ After all interviews were completed, a different group of Yawari and nearby Rukullakta residents was assembled in the community center for payouts for the Random Allocation and Dictator Local games. Each player's allocations were distributed to only one recipient, as indicated in the game instructions. However, multiple allocations were combined for each recipient to achieve a rough parity in distributions. At the end of the week, we traveled to Palimbe village, three hours' drive away, for randomized distribution of "distant" game allocations. Again, each player's allocations were distributed to only one recipient in Palimbe, but recipients in Palimbe all received allocations from several Yawari participants to even out distribution values.

2.3. Statistical Model

As detailed in our causal graph (Figure 2), our primary exposure is the joint effect of beliefs about Pachamama's proclivity for punishment and omniscience/knowledge breadth, and the outcomes are coin allocations to the distant co-religionist across two variants of the Random Allocation and Dictator Games. In all statistical models, we will adjust for all core confounders of age, sex, education, food security, number of children, and material security to block any potential back-door biasing paths through these variables. Although not confounders, all models will also hold game order and comprehension checks constant (the latter for RAG models only, as there were no comprehension checks for the DGs) to improve precision in our estimates.

Beyond this base model, as discussed above, Catholic beliefs and practices and Pachamama ritual practices may or may not also be confounders or mediators of the focal estimand. If these variables *are* confounders, then they need to be adjusted for in regression models to estimate our estimand without bias; if they are mediators, then they do *not* need to be adjusted for, as adjustment would induce bias; if there is reciprocal causation — meaning these variables are both confounders and mediators — then both adjusting and not adjusting for these variables will lead to bias. Given this uncertainty, we will compare models with all possible combinations of these variables as covariates in statistical models (i.e., four possible model combinations). If our estimate does not change based on these different assumptions, we can have more certainty that different causal assumptions regarding these variables do not impact our results. Alternatively, if

³ It should be noted that allocations added up to irregular values (totals for one or more Random Allocation Game $\neq$ \$30 and/or Dictator Game $\neq$ \$10) for a small number of participants (8/86). This could indicate cheating or stealing on the part of game players, or human error on the part of experimenters loading trays, or both. Participants were nevertheless given the allocations they had distributed to cups in each game.

our estimate is variable according to different assumptions, this uncertainty will be discussed when interpreting results. At the very least, the minimally and maximally adjusted models (risk of under-adjustment vs. overadjustment biases, respectively) likely bracket our causal effect of interest (Major-Smith, 2023).

We will use aggregated logistic regression models for the Random Allocation Games and ordinal models for Dictator Game allocations. Analyses will be run in Stan (Stan Development Team, 2026) using the “brms” package (Bürkner, 2017) for R version 4.4.1 (R Core Team, 2024) using four chains with 2,000 iterations (1,000 chains as warm-up) and weakly regularising priors. To ensure that chains are well-mixed and -powered, we will visually assess convergence between chains, $\hat{r}$ values (ideally 1.00), and effective sample sizes.

To estimate the marginal causal effect of supernatural punishment and omniscience beliefs on cooperation, we use *g*-computation methods (Hernán & Robins, 2020). As we do not know the causal relations between punishment and omniscience with any certainty — and hence cannot estimate the independent causal effects of these variables — we estimate the *joint* causal effect of these variables together; i.e., the marginal change in the outcome on the count scale from punishment and omniscience both as “0” to both as “1,” based on posterior predictions generated from our statistical models and the data.

We will initially treat beliefs in an additive fashion, followed by an interaction effect between punishment and omniscience. We will assess which model is a better fit using LOOIC (see Bendixen et al., 2023), although due to a relatively small sample size (just over 80), our analyses may be underpowered to detect said interaction effect (Blake & Gangestad, 2020). Example scripts demonstrating the proposed analyses on simulated data are available [here](#).

2.4. Sampling and Participants

Each participant was given \$20 as a show-up fee. Participants were recruited via word of mouth. The small size of Yawari and abutting Rukullakta settlements (particularly Rukullakta village, another of the 17 constituent villages of Rukullakta pueblo, which is geographically contiguous with Yawari) allowed word to spread quickly. The sample thus recruited was somewhat disproportionately female (73%). It is possible that the female bias in our sample reflects a higher likelihood for men to work outside the village, but that is speculative. Another equally possible explanation is that men in Yawari are simply less likely to participate in social events than women. The mean age of participants was 41.4 (SD = 17.3), indicating strong age diversity. As indicated in Section 1.2.2, participants reported growing about half their own food on average, suggesting limited market integration overall.

2.5. Pre-registration

As data has already been collected and audited, this is a pre-registration of our analysis plan, rather than for data collection and analysis. Plans for data collection can be found in the grant proposal, available [here](#). This knowledge of the data and ethnographic context has been useful to help understand some of the complexities of the data and help inform analysis decisions (e.g., the potentially complex relationship between the Christian God and Pacahamama, individual variation in market integration, and irregular game allocations). As also noted above, we initially developed core causal and statistical models for these site-specific analyses ([here](#)), which we then elaborated upon with the local context in mind for each of the site-specific papers. For full transparency, we note that descriptive statistics for key variables and regressions using this wider project core causal model were conducted in January 2026 for a project workshop, but the preliminary results of this analysis did not influence our causal model or corresponding statistical approach developed here.

Disclosure statement

The authors declare no conflicts of interest.

Authors' roles

CW and LC collected data, DM-S and BGP managed the project and contributed to analysis, and BGP contributed to the design. All authors contributed to writing this manuscript.

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Data availability statement

All data and code will be made available on the project GitHub website:
<https://github.com/bgpurzycki/GGSL-Project>.

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